Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Initial position

Meaning in my words How would you explain initial position to a friend?	**How can you tell?** At 0 s the cart is beside the +5 m mark. Where does its position graph begin?
Example or sketch The cart starts at +5 m and moves 2 m left. What two positions would you mark?	**Non-example and why** “The clock starts at zero, so the cart starts at zero.” What do the two zeros mean?

Bound

Meaning in my words How would you explain bound to a friend?	**How can you tell?** For 10 ± 2 km, what is the farthest allowed position on either side?
Example or sketch Mark the bounds for a forecast of 15 ± 3 km.	**Non-example and why** A student labels 2 km as the upper endpoint of 10 ± 2 km. What does the 2 mean?

Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Half-width

Meaning in my words How would you explain half-width to a friend?	**How can you tell?** A model allows 8 to 12 km. How far is either endpoint from 10 km?
Example or sketch How would you write a range centered on 20 km with half-width 3 km?	**Non-example and why** Someone calls 6 km the half-width of 17 to 23 km. What did they measure?

Assumption

Meaning in my words How would you explain assumption to a friend?	**How can you tell?** What must be true to predict another 2 m each second from a cart's earlier motion?
Example or sketch Write an assumption beside a straight-line forecast for XJ.	**Non-example and why** Is tomorrow's camera reading an assumption once we measure it?

Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

Draw a center tick and both endpoints. Show how starting-position and velocity bounds contribute to the final half-width.

Choose zero, direction, and scale.



My marks mean

How wide should we search?

05

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Read forecast bounds without inventing a probability.

CHOOSE ONE JOB TOGETHER

SURVEY ENGINEER ☐

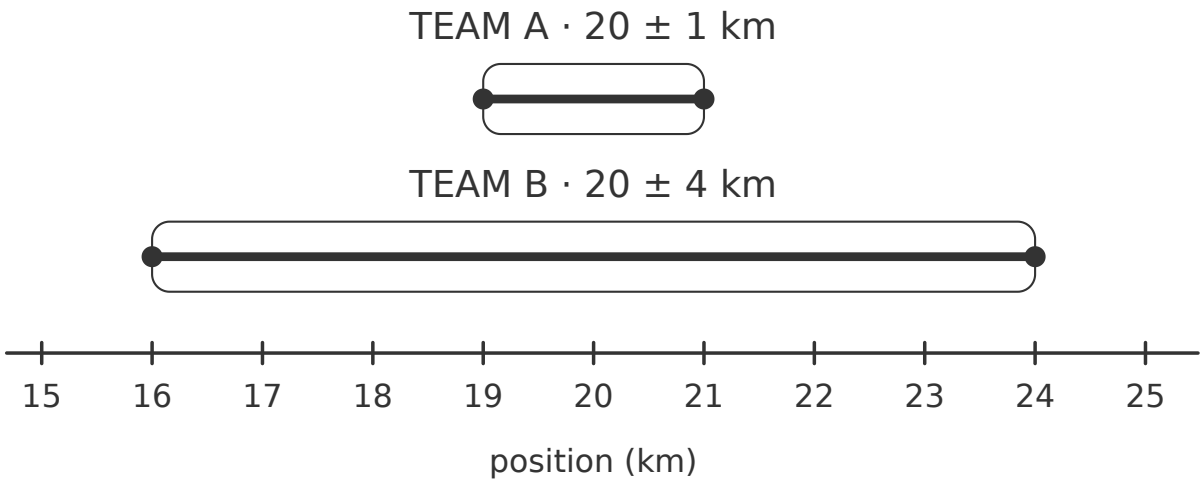
Label both search bands with their endpoints. Recommend which report needs a wider telescope search.

WEATHER-DESK PRESENTER ☐

Present both forecast bands. Explain which search is wider and what the bands do not tell us about impact.

Point to a clue. Keep your first marks; add the update using the color key.

Two XJ teams predict a center of **20 km**. Team A gives **±1 km**; team B gives **±4 km**. A telescope has one narrow field.



01 OUR FIRST CALL

What search area should the crew prepare for? Use both sets of endpoints.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

Check the editor's new label. What can you report from these bands?

Switch speakers. How is changing an input's uncertainty different from changing how you expect XJ to move?

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks